

GREELEY-WELD COUNTY AP, CO, USA

WMO: 724768

Lat: 40.436N

Lon: 104.632W

Elev: 4697

StdP: 12.37

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.7	0.3	-12.8	3.3	-6.8	-6.1	4.7	2.8	32.7	34.2	28.7	33.2	2.6	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.8	96.6	62.8	92.8	62.4	90.2	62.3	67.4	84.9	66.2	84.1	64.9	83.1	8.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	101.8	73.1	61.0	95.7	71.9	59.1	89.0	71.4	34.6	85.0	33.6	84.4	32.5	83.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.1	24.0	19.4	DB	-17.2	101.0	5.9	2.2	-21.4	102.6	-24.9	103.8	-28.2	105.0	-32.4	106.6	(4)
(5)			WB	-17.2	69.7	6.3	2.0	-21.7	71.1	-25.4	72.3	-28.9	73.4	-33.5	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	26.3	29.9	40.2	47.6	57.1	68.1	74.0	71.2	62.7	48.8	36.6	26.2		
	DBStd	18.78	10.93	11.16	9.32	8.15	7.93	6.49	4.16	4.28	7.24	8.53	9.48	10.73		
	HDD50	3053	734	565	318	138	29	1	0	0	7	121	403	737		
	HDD65	6471	1199	984	770	524	263	43	2	8	122	503	851	1202		
	CDD50	2744	0	1	13	64	248	543	745	656	388	83	3	0		
	CDD65	686	0	0	0	0	17	135	282	198	53	1	0	0		
	CDH74	10090	0	0	18	94	527	2089	3548	2526	1144	142	2	0		
	CDH80	4709	0	0	1	8	167	977	1876	1186	466	28	0	0		
	WSAvg	7.0	5.9	7.1	8.3	9.8	8.4	7.5	6.3	5.8	5.9	6.6	6.1	6.0		
	PrecAvg	13.4	0.3	0.4	0.7	1.7	2.2	1.6	1.7	1.7	1.2	1.1	0.5	0.4		
(15) Precipitation	PrecMax	18.2	0.9	0.9	3.6	6.8	6.1	3.9	3.7	5.7	4.1	3.3	1.6	1.3		
	PrecMin	7.5	0.0	0.0	0.0	0.1	0.6	0.1	0.3	0.1	0.0	0.3	0.1	0.1		
	PrecStd	2.8	0.3	0.2	0.9	1.3	1.2	1.0	0.9	1.2	0.9	0.7	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	66.2	76.5	81.3	90.5	99.2	100.0	96.9	94.0	83.7	72.1	60.7		
		MCWB	42.5	45.0	49.1	52.3	58.8	61.9	63.6	62.5	61.5	55.4	48.4	42.8		
	2%	DB	54.3	60.6	70.3	76.9	84.1	93.5	97.2	93.0	89.9	78.7	65.6	53.9		
		MCWB	39.7	42.5	47.2	51.0	57.0	61.0	63.5	62.6	60.0	53.2	47.1	39.8		
	5%	DB	48.0	54.2	65.5	72.5	80.7	90.3	93.3	90.5	85.8	72.6	60.8	47.8		
		MCWB	36.9	39.5	45.8	49.7	56.1	60.7	63.4	62.8	58.9	51.5	44.9	36.8		
	10%	DB	43.1	47.8	60.6	66.5	74.9	85.9	90.5	87.7	81.4	66.3	54.1	42.8		
		MCWB	33.9	36.5	43.3	47.9	54.7	60.2	63.7	62.9	58.4	49.3	41.5	33.5		
	0.4%	WB	43.8	46.0	50.6	54.8	63.1	67.6	69.5	68.6	65.4	57.1	49.8	43.7		
		MCDB	59.1	64.1	72.8	73.8	80.3	85.4	85.9	85.8	84.4	77.5	68.8	59.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.3	42.8	48.2	52.6	60.0	65.1	68.0	66.7	63.0	54.6	47.3	40.0		
		MCDB	52.9	58.9	68.1	71.7	76.7	83.6	85.3	84.1	82.3	73.7	64.6	52.4		
	5%	WB	37.3	39.9	46.0	50.6	58.0	63.4	66.8	65.5	61.1	52.6	44.8	36.9		
		MCDB	47.7	53.2	64.2	69.4	74.7	82.7	84.7	82.9	79.9	69.9	60.1	47.3		
	10%	WB	34.4	36.7	43.6	48.7	56.2	61.8	65.6	64.2	59.4	50.2	41.6	33.6		
		MCDB	42.5	47.1	58.9	65.8	72.4	81.7	84.2	81.7	76.4	65.4	53.5	41.5		
	5% DB	MDBR	27.7	27.4	30.5	28.8	27.8	30.3	31.8	31.4	31.8	30.0	28.6	26.7		
		MCDBR	35.4	37.5	41.5	41.1	37.4	38.6	38.0	37.4	40.5	41.2	38.7	34.7		
		MCWBR	23.1	22.3	22.0	19.8	15.2	13.2	12.1	12.9	15.6	20.1	22.3	22.7		
		MCWBR	33.6	35.1	39.3	36.6	32.3	33.0	31.6	31.9	34.2	36.3	37.6	33.0		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.252	0.273	0.304	0.327	0.333	0.361	0.352	0.315	0.275	0.257	0.246		
		taud	2.599	2.583	2.533	2.462	2.454	2.460	2.422	2.437	2.520	2.613	2.577	2.579		
	Ebn at Noon		303	313	314	308	301	298	288	288	295	300	293	292		
		Edh at Noon	21	25	30	34	35	35	36	35	30	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		767	1066	1472	1762	1980	2302	2222	1958	1652	1178	842	661		
	RadStd		36	68	106	120	162	115	90	56	83	78	48	45		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.88	+1.73	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page